

ln[]:= (* written by Si-Qiang Luo *)

ln[]:= (*< $\vec{\sigma}_i \cdot \vec{\sigma}_j$ >部分*)

configure = {"q", "q", "q̄", "q̄"}; (*夸克组成*)

(*自旋波函数*)

sw = {

(*11,2;2*)({1, "1111"}),

(*11,1;1*)($\{\frac{1}{2}, "1110"\}$, $\{\frac{1}{2}, "1101"\}$, $\{-\frac{1}{2}, "1011"\}$, $\{-\frac{1}{2}, "0111"\}$),

(*11,0;

0*)($\{\frac{1}{\sqrt{3}}, "1100"\}$, $\{\frac{1}{\sqrt{3}}, "0011"\}$, $\{-\frac{1}{2\sqrt{3}}, "1010"\}$,

$\{-\frac{1}{2\sqrt{3}}, "1001"\}$, $\{-\frac{1}{2\sqrt{3}}, "0110"\}$, $\{-\frac{1}{2\sqrt{3}}, "0101"\}$),

(*10,1;1*)($\{\frac{1}{\sqrt{2}}, "1110"\}$, $\{-\frac{1}{\sqrt{2}}, "1101"\}$),

(*01,1;1*)($\{\frac{1}{\sqrt{2}}, "1011"\}$, $\{-\frac{1}{\sqrt{2}}, "0111"\}$),

(*00,0;0*)($\{\frac{1}{2}, "1010"\}$, $\{-\frac{1}{2}, "1001"\}$, $\{-\frac{1}{2}, "0110"\}$, $\{\frac{1}{2}, "0101"\}$)

};

spin = {"11,2;2", "11,1;1", "11,0;0", "10,1;1", "01,1;1", "00,0;0"};

(*自旋基矢*)

swv[x_] := ($\frac{1}{0}$) /; x == "1"

swv[x_] := ($\frac{0}{1}$) /; x == "0"

Tswv[x_] := ($\frac{1}{0}$) /; x == "1"

Tswv[x_] := ($\frac{0}{1}$) /; x == "0"

(*Pauli矩阵*)

$\sigma[x_] := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ /; x == 1

$\sigma[x_] := \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$ /; x == 2

$\sigma[x_] := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ /; x == 3

$\sigma[x_] := 0$;

(*自旋求和 $\vec{\sigma}_i \cdot \vec{\sigma}_j$ *)

$\sigma_i \sigma_j$ ip_, i_, jp_, j_ := Sum[Tr[Tswv[ip]. $\sigma[x]$.swv[i]]*Tr[Tswv[jp]. $\sigma[x]$.swv[j]], {x, 1, 3}]

(*计算< $\vec{\sigma}_i \cdot \vec{\sigma}_j$ >*)

sisj = {};

For[ni = 1, ni ≤ Length[sw], ni++, For[nf = 1, nf ≤ Length[sw], nf++,

partsisj = {{spin[[ni]], spin[[nf]]}};

For[i = 1, i ≤ Length[configure] - 1, i++, For[j = i + 1, j ≤ Length[configure], j++,

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ss = 0;
For[pi = 1, pi ≤ Length[sw[[ni]], pi++, For[pf = 1, pf ≤ Length[sw[[nf]], pf++,
  qi = sw[[ni, pi, 2]]; qi = StringDrop[qi, {j}]; qi = StringDrop[qi, {i}];
  qf = sw[[nf, pf, 2]]; qf = StringDrop[qf, {j}]; qf = StringDrop[qf, {i}];
  If[qi == qf,
    ss = ss + sw[[ni, pi, 1]]*sw[[nf, pf, 1]]*σiσj[
      StringTake[sw[[nf, pf, 2]], {i}], StringTake[sw[[ni, pi, 2]], {i}],
      StringTake[sw[[nf, pf, 2]], {j}], StringTake[sw[[ni, pi, 2]], {j}]]
  ]
  ]];
partsisj = Join[partsisj, {ss}]
]];
sisj = Join[sisj, {partsisj}];
]];
sisj // MatrixForm

```

Out[]//MatrixForm=

$$\begin{pmatrix}
 \{11,2;2, 11,2;2\} & 1 & 1 & 1 & 1 & 1 & 1 \\
 \{11,2;2, 11,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,2;2, 11,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,2;2, 10,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,2;2, 01,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,2;2, 00,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,1;1, 11,2;2\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,1;1, 11,1;1\} & 1 & -1 & -1 & -1 & -1 & 1 \\
 \{11,1;1, 11,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,1;1, 10,1;1\} & 0 & \sqrt{2} & -\sqrt{2} & \sqrt{2} & -\sqrt{2} & 0 \\
 \{11,1;1, 01,1;1\} & 0 & -\sqrt{2} & -\sqrt{2} & \sqrt{2} & \sqrt{2} & 0 \\
 \{11,1;1, 00,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,0;0, 11,2;2\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,0;0, 11,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,0;0, 11,0;0\} & 1 & -2 & -2 & -2 & -2 & 1 \\
 \{11,0;0, 10,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,0;0, 01,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{11,0;0, 00,0;0\} & 0 & -\sqrt{3} & \sqrt{3} & \sqrt{3} & -\sqrt{3} & 0 \\
 \{10,1;1, 11,2;2\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{10,1;1, 11,1;1\} & 0 & \sqrt{2} & -\sqrt{2} & \sqrt{2} & -\sqrt{2} & 0 \\
 \{10,1;1, 11,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{10,1;1, 10,1;1\} & 1 & 0 & 0 & 0 & 0 & -3 \\
 \{10,1;1, 01,1;1\} & 0 & 1 & -1 & -1 & 1 & 0 \\
 \{10,1;1, 00,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{01,1;1, 11,2;2\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{01,1;1, 11,1;1\} & 0 & -\sqrt{2} & -\sqrt{2} & \sqrt{2} & \sqrt{2} & 0 \\
 \{01,1;1, 11,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{01,1;1, 10,1;1\} & 0 & 1 & -1 & -1 & 1 & 0 \\
 \{01,1;1, 01,1;1\} & -3 & 0 & 0 & 0 & 0 & 1 \\
 \{01,1;1, 00,0;0\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{00,0;0, 11,2;2\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{00,0;0, 11,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{00,0;0, 11,0;0\} & 0 & -\sqrt{3} & \sqrt{3} & \sqrt{3} & -\sqrt{3} & 0 \\
 \{00,0;0, 10,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{00,0;0, 01,1;1\} & 0 & 0 & 0 & 0 & 0 & 0 \\
 \{00,0;0, 00,0;0\} & -3 & 0 & 0 & 0 & 0 & -3
 \end{pmatrix}$$

In[]:= $\vec{\lambda}_i \cdot \vec{\lambda}_j$ 部分*)

configure = {"q", "q̄", "q̄", "q̄"}; (*夸克组成*)

(*色波函数*)

cw = {

```

(*6⊗6̄*){
  { $\frac{1}{\sqrt{6}}$ , "rrRR"}, { $\frac{1}{\sqrt{6}}$ , "ggGG"}, { $\frac{1}{\sqrt{6}}$ , "bbBB"},
  { $\frac{1}{2\sqrt{6}}$ , "rbBR"}, { $\frac{1}{2\sqrt{6}}$ , "brBR"}, { $\frac{1}{2\sqrt{6}}$ , "grGR"}, { $\frac{1}{2\sqrt{6}}$ , "rgGR"},
  { $\frac{1}{2\sqrt{6}}$ , "gbBG"}, { $\frac{1}{2\sqrt{6}}$ , "bgBG"}, { $\frac{1}{2\sqrt{6}}$ , "grRG"}, { $\frac{1}{2\sqrt{6}}$ , "rgRG"},
  { $\frac{1}{2\sqrt{6}}$ , "gbGB"}, { $\frac{1}{2\sqrt{6}}$ , "bgGB"}, { $\frac{1}{2\sqrt{6}}$ , "rbRB"}, { $\frac{1}{2\sqrt{6}}$ , "brRB"}},
(*3̄⊗3*){
  { $\frac{1}{2\sqrt{3}}$ , "rbBR"}, { $-\frac{1}{2\sqrt{3}}$ , "brBR"}, { $-\frac{1}{2\sqrt{3}}$ , "grGR"}, { $\frac{1}{2\sqrt{3}}$ , "rgGR"},
  { $\frac{1}{2\sqrt{3}}$ , "gbBG"}, { $-\frac{1}{2\sqrt{3}}$ , "bgBG"}, { $\frac{1}{2\sqrt{3}}$ , "grRG"}, { $-\frac{1}{2\sqrt{3}}$ , "rgRG"},
  { $-\frac{1}{2\sqrt{3}}$ , "gbGB"}, { $\frac{1}{2\sqrt{3}}$ , "bgGB"}, { $-\frac{1}{2\sqrt{3}}$ , "rbRB"}, { $\frac{1}{2\sqrt{3}}$ , "brRB"}},
};
color = {"6⊗6̄", "3̄⊗3"};
(*色基矢*)
cwv[x_] :=  $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$  /; x == "r"
cwv[x_] :=  $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$  /; x == "g"
cwv[x_] :=  $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$  /; x == "b"
Tcwv[x_] := (1 0 0) /; x == "r"
Tcwv[x_] := (0 1 0) /; x == "g"
Tcwv[x_] := (0 0 1) /; x == "b"
cwv[x_] :=  $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$  /; x == "R"
cwv[x_] :=  $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$  /; x == "G"
cwv[x_] :=  $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$  /; x == "B"
Tcwv[x_] := (1 0 0) /; x == "R"
Tcwv[x_] := (0 1 0) /; x == "G"

```

```
Tcww[x_] := (0 0 1) /; x == "B"
```

```
(*Gell-Mann矩阵*)
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```
 $\lambda[x_] := \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} /; x == 1$ 
```

```
 $\lambda[x_] := \begin{pmatrix} 0 & -I & 0 \\ I & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} /; x == 2$ 
```

```
 $\lambda[x_] := \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} /; x == 3$ 
```

```
 $\lambda[x_] := \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} /; x == 4$ 
```

```
 $\lambda[x_] := \begin{pmatrix} 0 & 0 & -I \\ 0 & 0 & 0 \\ I & 0 & 0 \end{pmatrix} /; x == 5$ 
```

```
 $\lambda[x_] := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} /; x == 6$ 
```

```
 $\lambda[x_] := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -I \\ 0 & I & 0 \end{pmatrix} /; x == 7$ 
```

```
 $\lambda[x_] := \frac{1}{\sqrt{3}} * \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} /; x == 8$ 
```

```
 $\lambda[x_] := 0;$ 
```

```
(*色求和 $\vec{\lambda}_i \cdot \vec{\lambda}_j$ *)
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```
 $\lambda_i \lambda_{ip\_}, i\_ , j_{p\_}, j\_ := \text{Sum}[\text{Tr}[\text{Tcww}[ip].\lambda[x].\text{cww}[i]] * \text{Tr}[\text{Tcww}[jp].\lambda[x].\text{cww}[j]], \{x, 1, 8\}];$ 
```

```
(*计算 $\vec{\lambda}_i \cdot \vec{\lambda}_j$ *)
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```
lilj = {};
```

```
For[ni = 1, ni ≤ Length[cw], ni++, For[nf = 1, nf ≤ Length[cw], nf++,
```

```
partlilj = {{color[ni], color[nf]}};
```

```
For[i = 1, i ≤ Length[configure] - 1, i++, For[j = i + 1, j ≤ Length[configure], j++,
```

```
ll = 0;
```

```
For[pi = 1, pi ≤ Length[cw[ni]], pi++, For[pf = 1, pf ≤ Length[cw[nf]], pf++,
```

```
qi = cw[ni, pi, 2]; qi = StringDrop[qi, {j}]; qi = StringDrop[qi, {i}];
```

```
qf = cw[nf, pf, 2]; qf = StringDrop[qf, {j}]; qf = StringDrop[qf, {i}];
```

```
If[qi == qf,
```

```
ll = ll + cw[ni, pi, 1] * cw[nf, pf, 1] * Which[
```

```
"q" == configure[i] && "q" == configure[j],  $\lambda_i \lambda_j$ 
```

```
StringTake[cw[nf, pf, 2], {i}], StringTake[cw[ni, pi, 2], {i}],
```

```
StringTake[cw[nf, pf, 2], {j}], StringTake[cw[ni, pi, 2], {j}]],
```

```
"q" == configure[i] && "q" == configure[j],  $-\lambda_i \lambda_j$ 
```

```
StringTake[cw[nf, pf, 2], {i}], StringTake[cw[ni, pi, 2], {i}],
```

```

      StringTake[cw[[ni, pi, 2]], {j}], StringTake[cw[[nf, pf, 2]], {j}]],
      "q" == configure[[i]] && "q" == configure[[j], -λ i λ j]
      StringTake[cw[[ni, pi, 2]], {i}], StringTake[cw[[nf, pf, 2]], {i}],
      StringTake[cw[[nf, pf, 2]], {j}], StringTake[cw[[ni, pi, 2]], {j}]],
      "q" == configure[[i]] && "q" == configure[[j], λ i λ j]
      StringTake[cw[[ni, pi, 2]], {i}], StringTake[cw[[nf, pf, 2]], {i}],
      StringTake[cw[[ni, pi, 2]], {j}], StringTake[cw[[nf, pf, 2]], {j}]]
    ]
  ]
];
partlilj = Join[partlilj, {ll}]
];
lilj = Join[lilj, {partlilj}];
]]
lilj // MatrixForm

```

Out[] // MatrixForm =

$$\begin{pmatrix}
 \{6\bar{6}, 6\bar{6}\} & \frac{4}{3} & -\frac{10}{3} & -\frac{10}{3} & -\frac{10}{3} & -\frac{10}{3} & \frac{4}{3} \\
 \{6\bar{6}, \bar{3}3\} & 0 & 2\sqrt{2} & -2\sqrt{2} & -2\sqrt{2} & 2\sqrt{2} & 0 \\
 \{\bar{3}3, 6\bar{6}\} & 0 & 2\sqrt{2} & -2\sqrt{2} & -2\sqrt{2} & 2\sqrt{2} & 0 \\
 \{\bar{3}3, \bar{3}3\} & -\frac{8}{3} & -\frac{4}{3} & -\frac{4}{3} & -\frac{4}{3} & -\frac{4}{3} & -\frac{8}{3}
 \end{pmatrix}$$

```

In[ ] := (* Color-spin basis *)
csw = {{1, {2, 2}},
  {{1/√2, {2, 4}}, {-1/√2, {2, 5}}},
  {1, {1, 2}},
  {{1/√2, {1, 4}}, {-1/√2, {1, 5}}},
  };

(* Calculate CMI *)
CMI = {};
For[ni = 1, ni ≤ Length[csw], ni++,
  onerow = {};
  For[nf = 1, nf ≤ Length[csw], nf++,
    CMIE = 0;
    For[pi = 1, pi ≤ Length[csw[[ni]], pi++, For[pf = 1, pf ≤ Length[csw[[nf]], pf++,
      For[i = 1, i ≤ Length[configure] - 1, i++, For[j = i + 1, j ≤ Length[configure], j++,
        CMIE = CMIE + csw[[ni, pi, 1]]*csw[[nf, pf, 1]]*lilj[[csw[[ni, pi, 2, 1]] - 1]*Length[cw] +
          csw[[nf, pf, 2, 1], (i - 1)*Length[configure] + j - (i + 1)/2 + 1]]*
          sisj[[csw[[ni, pi, 2, 2]] - 1]*Length[sw] + csw[[nf, pf, 2, 2], (i - 1)*Length[configure] +

```

```

      j -  $\frac{i(i+1)}{2} + 1$ ]]*ToExpression["C" <> ToString[i] <> ToString[j]]
    ]]
  ]];
  onerow = Join[onerow, {CMIe}];
];
CMI = Join[CMI, {onerow}]
];
Simplify[CMI]
(* Calculate Binding *)
Bind = {};
For[ni = 1, ni ≤ Length[csw], ni++,
  onerow = {};
  For[nf = 1, nf ≤ Length[csw], nf++,
    Binde = 0;
    For[pi = 1, pi ≤ Length[csw[[ni]]], pi++, For[pf = 1, pf ≤ Length[csw[[nf]]], pf++,
      If[csw[[ni, pi, 2, 2]] ≠ csw[[nf, pf, 2, 2]], Continue[]];
      For[i = 1, i ≤ Length[configure] - 1, i++, For[j = i + 1, j ≤ Length[configure], j++,
        Binde = Binde + csw[[ni, pi, 1]]*csw[[nf, pf, 1]]*
          lilj[(csw[[ni, pi, 2, 1]] - 1)*Length[cw] + csw[[nf, pf, 2, 1]], (i - 1)*Length[configure] +
            j -  $\frac{i(i+1)}{2} + 1$ ]]* $\left(-\frac{3}{8}\right)$ *ToExpression["B" <> ToString[i] <> ToString[j]]
      ]]
    ]];
  onerow = Join[onerow, {Binde}];
];
Bind = Join[Bind, {onerow}]
];
Simplify[Bind]
Out[ ] =  $\left\{ \left\{ -\frac{4}{3} (2 C_{12} - C_{13} - C_{14} - C_{23} - C_{24} + 2 C_{34}), -\frac{8}{3} (C_{13} - C_{24}), \right. \right.$ 
 $2 \sqrt{2} (-C_{13} + C_{14} + C_{23} - C_{24}), 4 \sqrt{2} (C_{13} - C_{24}) \left. \right\}, \left\{ -\frac{8}{3} (C_{13} - C_{24}), \right.$ 
 $\frac{4}{3} (2 C_{12} + C_{13} - C_{14} - C_{23} + C_{24} + 2 C_{34}), 4 \sqrt{2} (C_{13} - C_{24}), -2 \sqrt{2} (C_{13} + C_{14} + C_{23} + C_{24}) \left. \right\},$ 
 $\left\{ 2 \sqrt{2} (-C_{13} + C_{14} + C_{23} - C_{24}), 4 \sqrt{2} (C_{13} - C_{24}), \frac{2}{3} (2 C_{12} + 5 C_{13} + 5 C_{14} + 5 C_{23} + 5 C_{24} + 2 C_{34}), \right.$ 
 $-\frac{20}{3} (C_{13} - C_{24}) \left. \right\}, \left\{ 4 \sqrt{2} (C_{13} - C_{24}), -2 \sqrt{2} (C_{13} + C_{14} + C_{23} + C_{24}), \right.$ 
 $-\frac{20}{3} (C_{13} - C_{24}), -\frac{2}{3} (2 C_{12} - 5 C_{13} + 5 C_{14} + 5 C_{23} - 5 C_{24} + 2 C_{34}) \left. \right\} \}$ 

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$$\begin{aligned}
 Out[+]= & \left\{ \left\{ -\frac{1}{2} (2 B_{12} + B_{13} + B_{14} + B_{23} + B_{24} + 2 B_{34}), 0, -\frac{3 (B_{13} - B_{14} - B_{23} + B_{24})}{2 \sqrt{2}}, 0 \right\}, \right. \\
 & \left\{ 0, \frac{1}{2} (2 B_{12} + B_{13} + B_{14} + B_{23} + B_{24} + 2 B_{34}), 0, -\frac{3 (B_{13} - B_{14} - B_{23} + B_{24})}{2 \sqrt{2}} \right\}, \\
 & \left\{ -\frac{3 (B_{13} - B_{14} - B_{23} + B_{24})}{2 \sqrt{2}}, 0, \frac{1}{4} (-2 B_{12} + 5 B_{13} + 5 B_{14} + 5 B_{23} + 5 B_{24} - 2 B_{34}), 0 \right\}, \\
 & \left. \left\{ 0, -\frac{3 (B_{13} - B_{14} - B_{23} + B_{24})}{2 \sqrt{2}}, 0, \frac{1}{4} (-2 B_{12} + 5 B_{13} + 5 B_{14} + 5 B_{23} + 5 B_{24} - 2 B_{34}) \right\} \right\}
 \end{aligned}$$